

HEATMAP NODE · REQUIREMENTS

HeatMap Node requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/03-requirements.md @ d91cd4d		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3: R13 redefined to the sensor head per HMN-DDR-001 D1; status table replaced by the results of HMN-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

HeatMap Node requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be revised after co-design sessions (see HMN-PRB-001). Version 0.3 redefines R13 to cover the sensor head only, under HMN-DDR-001 D1 (adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review). No other target changes. Status is from the TRL 3 calculations in HMN-CAL-001; nothing has been tested.

Six requirements are **not met** on paper (R1, R4, R8, R9, R13 and R15), three are **at risk** (R5, R10 and R14), four are met on paper and two are met by design. See Table 2.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Air temperature accuracy in the street	± 0.5 °C against a reference aspirated thermometer, in full sun at wind speeds of 1 m/s or more; ± 1.0 °C below 1 m/s	Shield error estimate; later side-by-side field comparison
R2	Relative humidity accuracy	± 3 %RH from 20 to 80 %RH at 15 to 40 °C	Datasheet; CalRig salt fixed points
R3	Globe temperature	Standard 150 mm black globe (emissivity about 0.95); probe ± 0.3 °C after calibration; 90 % response within 30 min	Calibration in CalRig; response estimate
R4	Wind speed	0.5 to 20 m/s; ± 0.5 m/s or ± 10 %, whichever is larger	Datasheet of the chosen anemometer; later wind comparison
R5	Derived heat stress	Mean radiant temperature and outdoor WBGT computed per ISO 7243 definitions with the Liljegren method; WBGT within ± 1.0 °C of a reference WBGT meter	Calculation note; later collocation with a reference meter
R6	Reporting	15 min means of all channels by default (adjustable 5 to 60 min); data at the gateway within 30 min; 7 days stored on the node for resend	FieldNode firmware design; airtime calculation
R7	Power autonomy	No mains; 5 days without sun at the default reporting interval	FieldNode energy budget plus the sensor load in HMN-PRC-001
R8	Outdoor survival	Electronics IP65; operate at -20 to +60 °C at the node; UV-stable plastics; sensors survive driving rain and dust	Design review; datasheets
R9	Measurement height	Sensors at pedestrian height (1.1 to 2.0 m)	Mounting review with pole owners
R10	Fit to existing poles	Stainless band clamps for 60 to 200 mm round poles; no drilling; fitted from a ladder or lift by two people in 30 min or less	Design review; later timed trial
R11	Wind survival	No failure at a 35 m/s gust; arm deflection under 10 mm at 20 m/s	Wind load calculation
R12	Privacy	Environmental data only: no camera, microphone, Wi-Fi or Bluetooth scanning, or any personal data	Design review of hardware and firmware
R13	Cost	Sensor head (all parts except the FieldNode core) \$120 or less, the <code>budget_usd</code> in <code>project.yaml</code> . The FieldNode core is costed in the FieldNode repo; the full-node cost is reported for information	Priced BOM
R14	Traceable data	Each node carries a calibration record from CalRig; readings published in an open, documented format with node location and height	Documentation review
R15	Mass and loading on the pole	4 kg or less for the complete node; no part larger than 300 mm across other than the arm	Mass estimate

Table 2. Status from HMN-CAL-001 (TRL 3 calculations).

ID	Value (HMN-CAL-001)	Status
R1	1.04 °C at 1 m/s and 2.09 °C at 0.5 m/s in full sun with the naturally ventilated shield; 0.09 °C with a fan (option)	Not met (assumption-sensitive)
R2	SHT45 ±1.0 %RH typical	Met by design
R3	Probe ±0.18 °C (±0.26 °C with a maximum-tolerance reference); 90 % response about 12 min	Met on paper
R4	Anemometer start-up about 0.8 m/s (typical, unverified); zero read below it biases MRT by -8.2 °C at a true 0.5 m/s	Not met
R5	Input error ±0.59 °C RSS, ±0.88 °C worst case; +0.9 °C in calm air below start-up; Liljegren model error unknown	At risk
R6	20 B payload; 23.7 s/day at SF9; 7 days in 21.5 kB; 16 min latency for 99 % of intervals	Met on paper
R7	Sensor load 0.116 mW, 0.12 % of FieldNode's 100 mW design allowance	Met on paper
R8	FieldNode interior 78.3 °C at 50 °C air and 88.3 °C at 60 °C against a 70 °C rating (FND-CAL-001); sensor head parts within ratings	Not met (FieldNode core)
R9	Sensors at 2.67 to 2.71 m; pedestrian-height air 0.2 to 0.8 °C warmer in strong sun	Not met
R10	120° V-saddle fits 60 to 200 mm; 37 min estimated	At risk (time)
R11	Arm factor 13.8 on yield; 0.29 mm at 20 m/s; clamp twist factor 2.6	Met on paper (preload assumed)
R12	No imaging, audio or radio scanning parts; environmental payload only	Met by design
R13	Sensor head \$130.00; full node \$256.00	Not met (sensor head \$10 over)
R14	Probe calibrated before fitting fits a CalRig bay; the assembled 150 mm globe does not; payload format defined	At risk
R15	4.68 kg complete node (FieldNode 2.41 kg); largest part 290 mm	Not met (mass)

■ Assumptions

- The FieldNode core performs as stated in FND-CAL-001 (100 mW design sensor allowance, 5.75 days without sun at that load; 2.41 kg; \$126.00).
- WBGT is derived, not measured with a wetted wick, because a natural wet-bulb sensor needs a water reservoir that would need refilling at every node.
- Pole owners permit clamped attachments at the proposed height.